# Spotify Recommendations: Machine Learning

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# Data for Learning Models

**Artist-Sourced Metadata**

- Genre and Sub-genre
- Song Moods
- Instruments and Styles

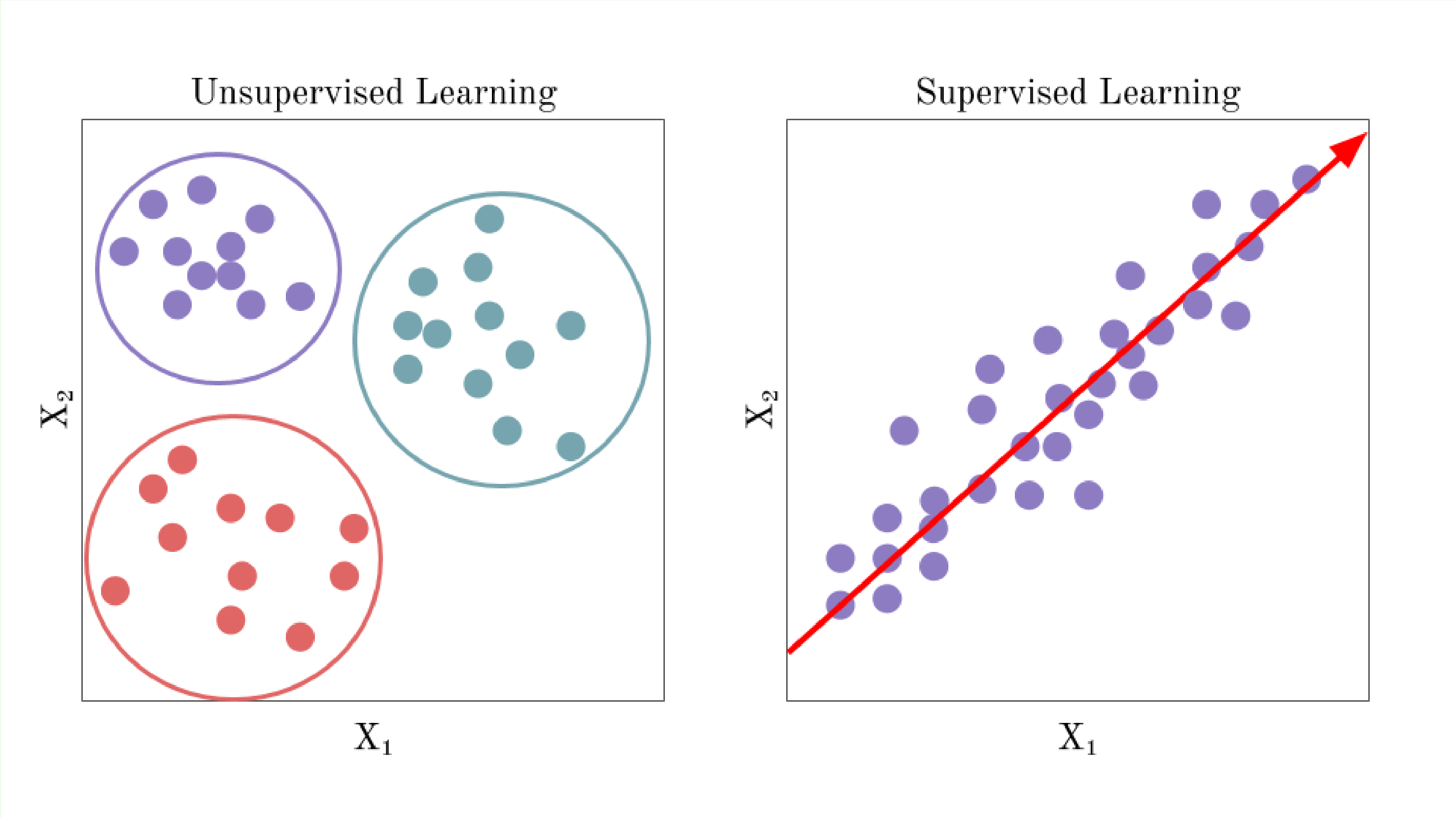
**User Engagement Metrics**

- Skips and Saves (Likes/Dislikes)
- Playlist Additions
- Listening Patterns

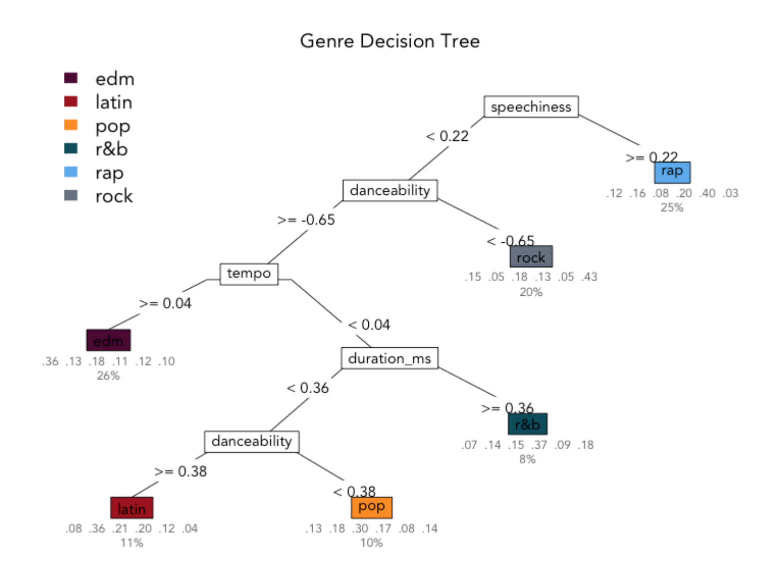
**Audio Feature Analysis**

- Acousticness
- Danceability
- Energy
- *Valence*
- Tempo


# Supervised and Unsupervised Learning Techniques



# Supervised Learning Techniques in Spotify Recommendations: <u>Decision Trees</u>




**What is it?**

*Splits

*Node

*Leaf Node

**How Do They Work?**

*Root Node = Specific Feature

*Model Splits Data

*Separates Data into Distinct Classes

*Final Decision at leaf node

**How Does Spotify Use a Decision Tree**

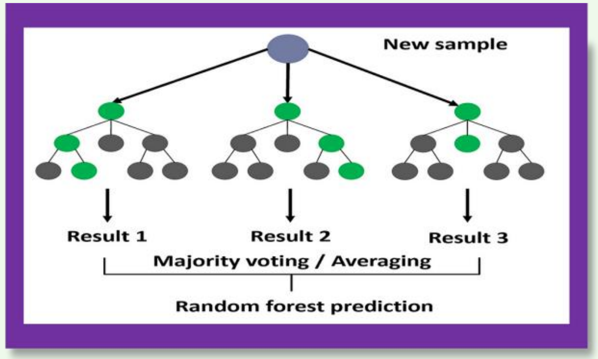
*Training the Tree*

*Song = Features, Label = Like/Dislike

*Making Predictions*

*New Song = decision tree follows a path

ex. User has liked high-tempo rock songs in the past, tree may first check if a song is rock. If it is, it will then look at tempo.




**Why Do They Use It?**

*Interpretability and Transparency

*Ability to Handle Missing Data

*Handling Non-Linear Relationships

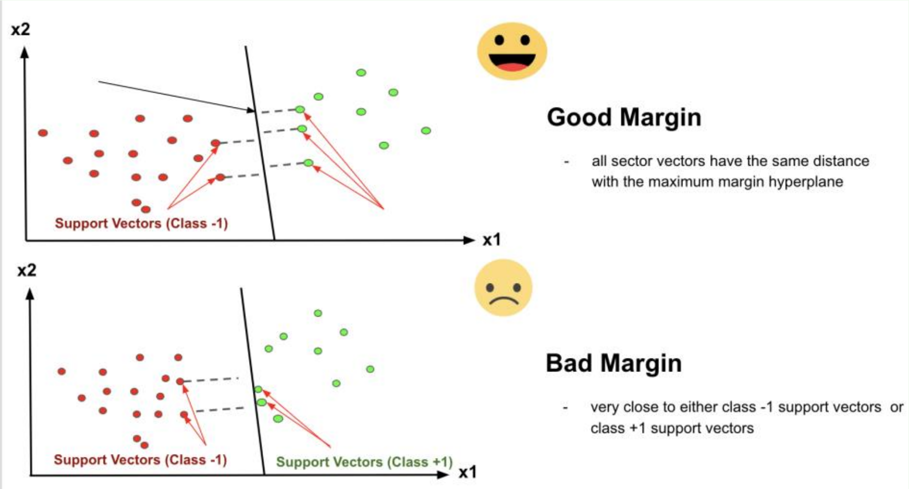
*Efficiency in Real-Time Recommendations

*Etc.

**Limitations:**

*Decision trees can sometimes be overly specific, but ensemble methods like Random Forests (using multiple decision trees) help make the model more robust.

# Supervised Learning Techniques in Spotify Recommendations: <u>Support Vector Machines</u>



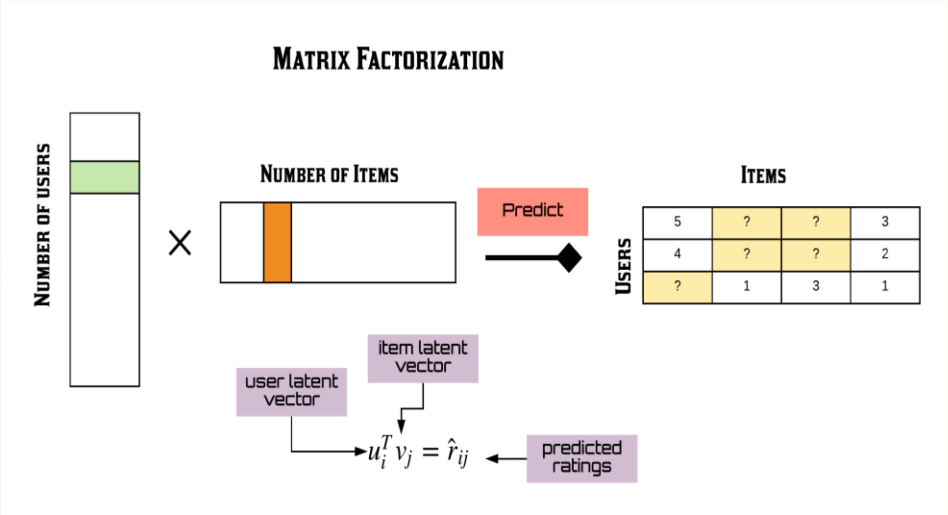

**What is it?**
*Classifies Data with separation
**Good Margin*
maximum separation
**Bad Margin*
less separation
higher risk of misclassification

**How Do They Work?**
*SVM aims to find a hyperplane
*Points "support" the hyperplane and determine its position
*SVM uses hyperplane to classify new data points based to a side

**How Does Spotify Use a Decision Tree**
*Collects data on songs interacted with
*Songs the user likes/dislikes are labeled
*model learns how to separate two types of similar songs
*SVM creates a decision boundary and maximizes margin

**Why Do They Use It?**
*Handling High-Dimensional Data
*Adaptability to Complex Data
*Highly Accurate for Distinct Preference Patterns
*Personalized Recommendations

# Unsupervised Learning Technique for Spotify Recommendations: <u>Matrix Factorization</u>




**What is it?**
*A way of doing Collaborative Filtering
*Breaks down large matrix of user-item interactions
*Finding Latent Factors

**Why Do They Use It?**
*Discovery of Hidden Patterns
*No Need for Detailed Features
*Effective for Sparse Data

**How Does Spotify Use SVM**
*Each user and song pair is represented in a matrix.
*ex. User A likes Song X but disliked Song Y, these are stored as entries in the matrix.*
*Matrix factorization algorithm
*Matching Latent Factors
*Recommends songs that similar users enjoy

# Spotify Mood Playlist Algorithms

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